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3D POINT CLOUD ACQUISITION USING AN RGB CAMERA MOUNTED ON A ROBOT ARM

Summer Internship Progress Presentation

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Internship Context

Summary Internship Framework

- **Student:** Aymen Turki
- **Supervisor:** Faouzi Alaya Cheikh
- **Host Institution:** Norwegian University of Science and Technology (NTNU)
- **Research Group:** Intelligent Systems and Analytics (ISA), Department of Computer Science
- **Core Task:** AI-driven 3D point cloud acquisition for robotic plant phenotyping
- **Integration:** Contribution to the **PheNo** plant phenotyping infrastructure

Introduction to Plant Phenotyping

What is Plant Phenotyping?

- The quantitative analysis of structural, physiological, and anatomical plant characteristics (phenotypes).
- Essential for precision agriculture, crop breeding, stress analysis, and growth monitoring.

The Need for Automation

- Traditional manual methods are invasive, slow, and prone to human error.
- Automated vision systems enable continuous, high-throughput monitoring without damaging plants.

Why 3D Point Cloud Acquisition?

2D vs. 3D Vision

- **2D Limitations:** Struggles with leaf overlap, self-occlusion, and lack of depth geometry.
- **3D Advantage:** Captures true 3D spatial structure, volume, and leaf orientation.

Monocular RGB + AI Approach

- Standard monocular cameras are lightweight and cost-effective compared to heavy 3D sensors (LiDAR/Depth cams).
- Advanced AI models reconstruct dense 3D point clouds directly from monocular RGB frames.

Project Overview & Task Objectives

- **Topic:** 3D point cloud acquisition using a standard monocular RGB camera mounted on a robot arm relies on AI.
- **Primary Goal:** Develop an automated solution for 3D plant reconstruction and phenotyping feature extraction.

Main Objectives

1. Create 3D models of plants using monocular RGB images.
2. Calculate topological and geometric features for plant phenotyping.
3. Integrate the final 3D plant model as a core part of the **PheNo** infrastructure.

Robot Platform: JetCobot 7-Axis Arm

Hardware Overview

- **JetCobot:** 7-axis collaborative robotic arm by Elephant Robotics, built for precise, flexible manipulation.
- **Onboard Compute:** Powered by an NVIDIA Jetson Orin Nano, enabling edge AI inference directly on the arm.
- **End Effector:** Integrated gripper with a monocular RGB camera mounted at the wrist for close-range imaging.
- **Connectivity:** Wireless module for remote monitoring and control.

JetCobot 7-Axis Robotic Arm

(Elephant Robotics)

ROS Jazzy Simulation & Motion Planning

ROS Jazzy Integration

- **Middleware:** ROS 2 Jazzy Jalisco used as the core communication and control framework.
- **Motion Planning:** MoveIt2 configured for the JetCobot kinematic chain to plan collision-free trajectories.
- **Simulation:** Gazebo used to simulate the arm and environment

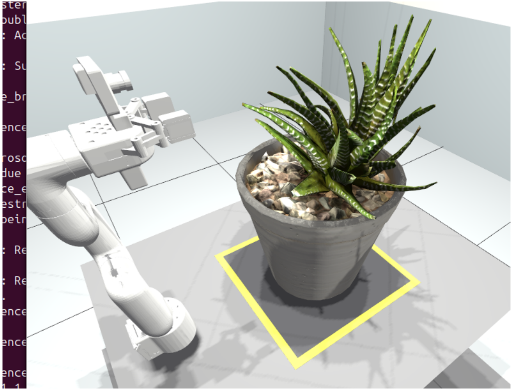
Virtual Plant Environment

- **3D Assets:** Plant models created in Blender and imported into the Gazebo simulation.
- **Scene Setup:** Plants positioned within the arm's workspace to emulate real scanning conditions.
- **Sim-to-Real:** Simulation environment used to validate trajectories prior to deployment on hardware.

Data Capture Pipeline

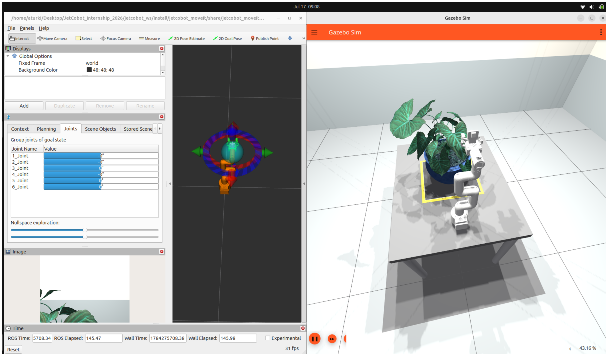
Camera & Kinematics

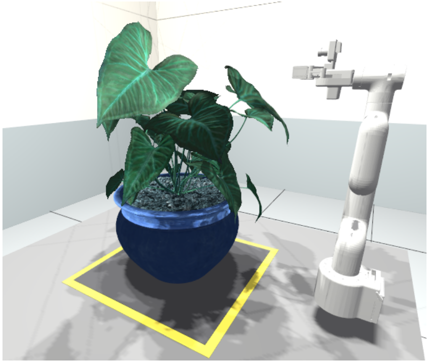
- **Camera Topic:** A dedicated ROS topic publishes RGB images from the wrist-mounted camera in real time.
- **Forward Kinematics:** End-effector pose computed via forward kinematics and logged alongside each frame.
- **Synchronization:** Each image is timestamped and paired with its corresponding arm pose.



Test capture: JetCobot arm scanning a plant target

Robot Platform and Plant Environment

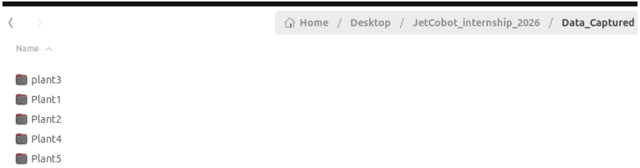




Data Capture Pipeline

Dataset Construction

- **Storage:** Captured RGB images saved locally, structured for downstream processing.
- **Pose Logging:** Corresponding forward-kinematics poses saved with each image to build an image–pose dataset.

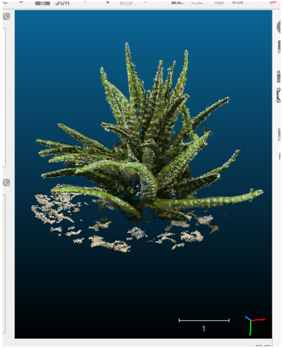


Dataset generation: synchronized image–pose acquisition

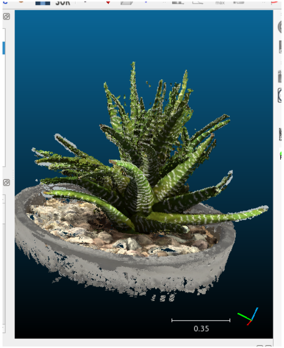
3D Reconstruction: COLMAP vs. MASt3R

Methods

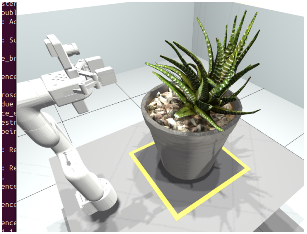
- **COLMAP:** Classical SfM baseline.
- **MASt3R:** Deep learning-based reconstruction & matching.



COLMAP Point Cloud



MASt3R Point Cloud



Ground Truth Plant

NeRF and Gaussian Splatting

NeRF

- **Neural Radiance Fields (NeRF):** learns a continuous 3D representation of a scene from multiple posed RGB images.
- Produces high-quality novel view synthesis through neural volume rendering.
- Two pose estimation pipelines were evaluated:
  - NeRF (COLMAP)
  - NeRF (MASt3R)

Gaussian Splatting

- **3D Gaussian Splatting (GS):** represents the scene using thousands of anisotropic 3D Gaussians.
- Enables real-time rendering while preserving high visual quality.
- Initialized using camera poses and sparse reconstruction from:
  - Gaussian Splatting (COLMAP)

3D Reconstruction Evaluation

Evaluation Metrics

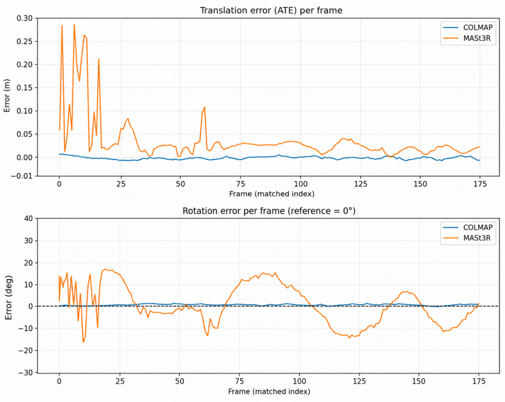
- **PSNR:** Higher values indicate better reconstruction fidelity.
- **SSIM:** Higher values indicate better structural similarity.
- **LPIPS:** Lower values indicate better perceptual quality.
- **Speed (fps):** Higher values indicate faster rendering.

Metric	NeRF (COLMAP)	NeRF (MASt3R)	GS (COLMAP)	Winner
PSNR ↑	19.93	14.17	**19**	NeRF (COLMAP)
SSIM ↑	0.79	0.55	**0.85**	GS (COLMAP)
LPIPS ↓	0.16	0.42	**0.11**	GS (COLMAP)
Speed (fps)	0.096	0.095	**0.085**	NeRF (COLMAP)

Camera Pose Accuracy: COLMAP vs. MASt3R

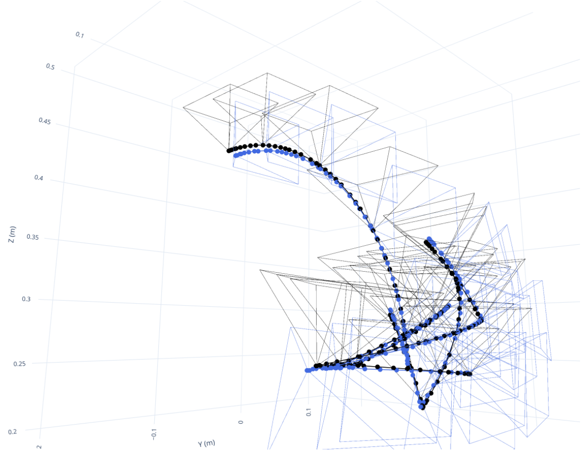
Evaluation

- **Translation Error:** Absolute Trajectory Error (ATE) RMSE.
- **Rotation Error:** Relative orientation consistency across frames.
- Comparison performed against the Blender ground-truth camera trajectory.



Translation and rotation error evolution over the camera trajectory

Camera Trajectory: COLMAP vs. Ground Truth



Reading the Plot

- **Black:** ground-truth camera path, from the robot's forward kinematics during the scan.
- **Blue:** camera poses estimated by **COLMAP**.

Observation

COLMAP's estimated path closely tracks the true trajectory, staying tightly aligned with the robot's actual arc throughout the scan.

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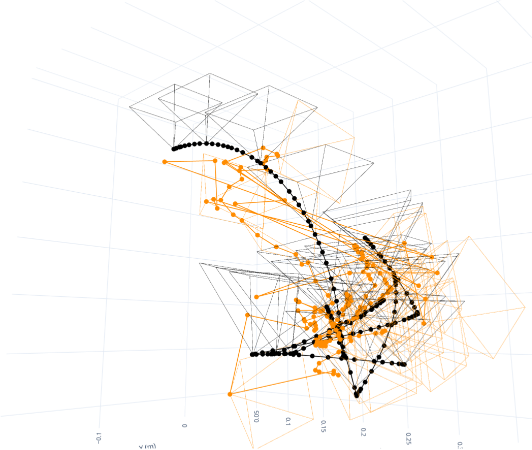
Camera Trajectory: MASt3R vs. Ground Truth



Reading the Plot

- **Black:** ground-truth camera path, from the robot's forward kinematics during the scan.
- **Orange:** camera poses estimated by **MASt3R**.

Observation

MASt3R's estimated path deviates sharply from the true hemispherical scan — large jumps and tangled loops, especially at the start of the trajectory.

3D Plant Phenotyping Pipeline

From Reconstruction to Phenotypic Traits

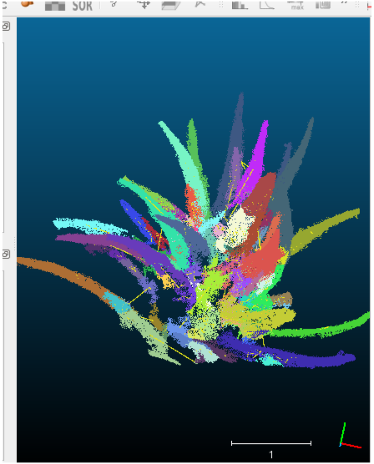
3D Reconstruction → Plant Segmentation → Skeleton Extraction → Trait Quantification

Step 1
3D Plant Segmentation

Step 2
Skeletonization & Graph Extraction

Step 3
Phenotypic Trait Quantification

Step 1: 3D Plant Segmentation



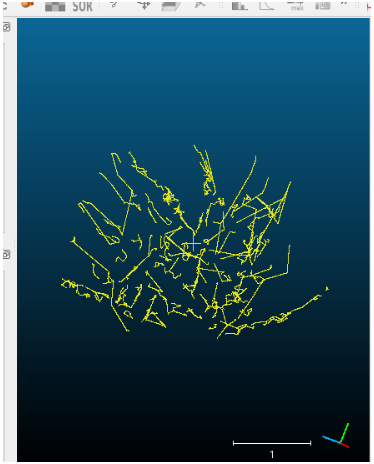
Process

- **Background Removal:** pot, table, and lab background filtered from the point cloud.
- **Organ Segmentation:** individual leaves extracted as separate point sets.

Result

- **90 leaves** segmented from the reconstructed plant.
- Each leaf stored independently for further analysis.

Step 2: Skeletonization & Graph Extraction



Process

- **Curve Skeleton:** leaf point clouds reduced into a 1D graph representation.
- **Graph Structure:** captures leaf axis and geometric organization.

Result

- Skeleton representation generated for segmented leaves.
- Graphs used for downstream trait extraction.

Step 3: Phenotypic Trait Quantification

Extracted Traits (per leaf)

- **Geometry:** length, width, thickness, orientation, height.
- **Topology:** skeleton nodes, edges, and structural properties.
- **Health:** RGB/HSV color, healthy tissue estimation, surface roughness.

Metric (90 leaves)	Value
Average leaf length	0.1 m
Longest leaf	0.35 m
Average leaf width	0.028 m
Average healthy tissue	93%

Future Research Direction

Future Research

- **AI-Based Plant Topology Reconstruction:** Generate synthetic Blender datasets with ground-truth graphs, reconstruct plants using COLMAP/MASt3R, and train a neural network to infer biological topology for automatic plant phenotyping.
- **Generalization:** Evaluate the complete pipeline on different plant species and architectures to assess robustness and scalability.
- **Adaptive View Planning:** Replace the fixed scanning sequence with an intelligent Best Next View strategy for autonomous and more efficient 3D reconstruction.